Unit 4

Shocking the Equilibrium

Objectives of the Study

After studying this unit, the student should be able:

- to know how the equilibrium is shocked and its effects upon demand curve and supply curve

Shocking the Equilibrium

Once an equilibrium is achieved, it can persist indefinitely because no one applies pressure to change the price. The equilibrium changes only if a shock occurs that shifts the demand curve or the supply curve. These curves shift if one of the variables we were holding constant changes. If tastes, income, government policies, or costs of production change, the demand curve or the supply curve or both shifts, and the equilibrium changes. That is, shocking the equilibrium refers to a situation where an external event or change (a "shock") disrupts the existing balance in a market or economy, causing shifts in supply, demand, or both, and ultimately leading to a new equilibrium point. This shock can be positive (increasing quantity demanded or supplied) or negative (decreasing quantity demanded or supplied).

A shock is an unexpected event or change that disrupts the established equilibrium. It can originate from various sources, such as changes in consumer preferences, technological advancements, natural disasters, or government policies and can be either positive (beneficial) or negative (harmful) for the market.

Effects of a Shift in the Demand Curve

Demand shocks are also commonly perceived to come about because of changes in consumer preferences, but they can also be linked to changes in other factors of demand like the price of complements and substitutes. For example, if there is a sudden increase in consumer income, this could lead to an increase in demand for certain goods or services. If suppliers are unable to adjust their production levels quickly enough to meet this increased demand, a shortage may

occur, disrupting the equilibrium. Conversely, a sudden decrease in demand, perhaps due to a change in consumer preferences, could lead to a surplus if suppliers are unable to reduce their production levels in time. Demand shocks can be negative or positive.

a) Negative Demand Shocks

These cause less quantity of goods to be consumed, and those consumers still in the market pay a lower price for the good. An example of this would be if a medical journal reported that a widely used prescription drug appreciably increases your chances of cancer. Then there would be a sharp shift in demand with less goods being consumed at a lower price.

b) Positive Demand Shock

Conversely, this type of shock can cause more goods to be consumed at a higher price. Consider the combined effects of two simultaneous events upon the demand of two complements, fish and tartar sauce: a new nutritional study conclusively touts the many health benefits of eating fish, and there are commercial fishing efficiency advances that make fish in inexpensive. More people will want to eat fish, leading them to buy more tartar sauce even though it's now at a higher price.

Demand shocks can be explained with the help of figure 4.1

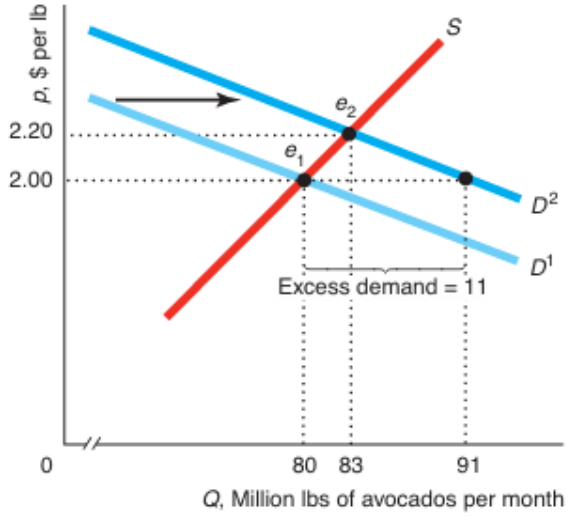


Figure 4.1: Demand Shocks

Suppose that the price of fresh tomatoes increases by $55 per lb, so consumers substitute avocados for tomatoes. As a result, the demand curve for avocados shifts outward from D_1 to D_2. At any given price, consumers want more avocados than they did before the price of tomatoes rose. In particular, at the original equilibrium price of avocados of $2, consumers now want to buy 91 million lbs of avocados per month. At that price, however, suppliers still want to sell only 80 million lbs. As a result, excess demand is 11 million lbs. Market pressures drive the price up until it reaches a new equilibrium at $2.20. At that price, firms want to sell 83 million lbs and consumers want to buy 83 million lbs, the new equilibrium quantity. Thus, the equilibrium moves from e_1 to e_2 as a result of the increase in the price of tomatoes. Both the equilibrium price and the equilibrium quantity of avocados rise as a result of the outward shift of the avocado demand curve. Here the increase in the price of tomatoes causes a shift of the demand curve, which in turn causes a movement along the supply curve.

Effects of a Shift in the Supply Curve

a shock to supply occurs when there is an unexpected change in the factors affecting supply, such as production costs, technology, or the availability of resources. For instance, a sudden increase in the cost of raw materials could lead to a decrease in supply as producers are unable or unwilling to produce as much at the same price. If demand remains constant, this could lead to a shortage, disrupting the equilibrium. On the other hand, a sudden increase in supply, perhaps due to a technological advancement, could lead to a surplus if demand remains constant.

The supply of goods and services are often the ones who face shocks, though they can affect producers and consumers alike.

a) Negative Supply Shock

Negative supply shocks cause the quantity supplied to be rapidly reduced, and the price to increase quickly until a new equilibrium is reached. A good example of this would be any natural disaster or other unanticipated event that disrupts the production process and/or supply-chain. An instance of this would be Hurricane Katrina's detrimental affects upon the oil and gasoline industry: oil rigs, refine plants, and pipelines were either shut down or taken off line. A hypothetical example of this could be if a key resource input of a firm's production process

was found to have a much more valuable application in another use, then the cost of the input would rise. For instance, if a pipe-fitter's firm's production process used a metal that the microchip industry found a novel use for, the price of that metal would increase. Thus, the number of pipes supplied would fall and their price would increase.

b) Positive Supply Shock

These usually come in the form of overnight technological advances that quickly improve the productivity of labour and the return of capital. These improvements cause the quantity supplied to increase and the price to fall. For instance, this age of computers and robots has represented an unprecedented increase in productivity that goods can be mass produced on a massive but relatively inexpensive scale. It could also occur if a new, cheaper substitute for an expensive production input is found. For instance, the discovery of a seam of low-sulfar coal out in Power basin, Wyoming.

Supply shocks can be explained with the help of figure 4.2

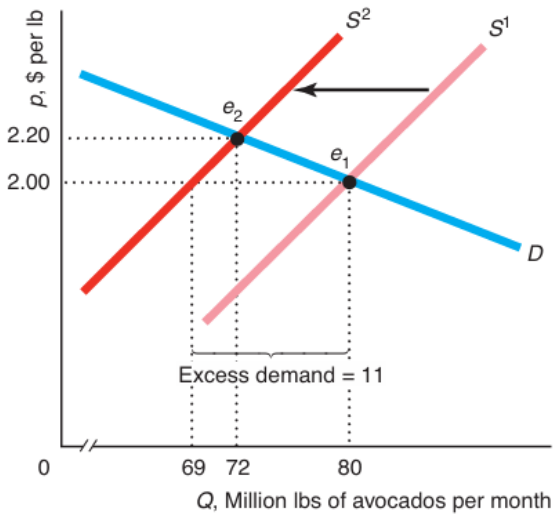


Figure 4.2: Supply Shocks

Now suppose that the price of tomatoes stays constant at its original level but the price of fertilizer rises by $ 0.55 per lb. It is now more expensive to produce avocados because the price of an important input, fertilizer, has increased. As a result, the supply curve for avocados shifts to the left from S1 to S2 in Figure 1.24. At any given price, producers want to supply fewer avocados than they did before the price of fertilizer increased. At the original equilibrium price for avocados of $2 per lb, consumers still want 80 million lbs, but producers are now willing

to supply only 69 million lbs, so excess demand is 11 million lbs. Market pressure forces the price of avocados up until it reaches a new equilibrium at e_2, where the equilibrium price is \$2.20 and the equilibrium quantity is 72. The increase in the price of fertilizer causes the equilibrium price to rise but the equilibrium quantity to fall. Here a shift of the supply curve results in a movement along the demand curve. In summary, a change in an underlying factor, such as the price of a substitute or the price of an input, shifts the demand curve or the supply curve. As a result of a shift in the demand or supply curve, the equilibrium changes. To describe the effect of this change, we compare the original equilibrium price and quantity to the new equilibrium values.

Multiple Choice Questions

1. What occurs when a shock disrupts the existing market equilibrium?

   a) The supply curve shifts to the right.

   b) The equilibrium remains unchanged.

   c) The demand curve shifts, leading to a new equilibrium.

   d) The price of all goods decreases.

   Answer: (c)

2. A positive demand shock typically results in which of the following outcomes?

   a) An increase in the quantity of goods consumed at a lower price.

   b) A decrease in consumer preferences for certain goods.

   c) An increase in the quantity of goods consumed at a higher price.

   d) A surplus of goods in the market.

   Answer: (c)

3. Which of the following best describes a negative supply shock?

   a) An unexpected event that decreases production costs, leading to an increase in supply.

   b) An event that increases the quantity supplied and decreases prices.

   c) A situation where production is disrupted, causing a decrease in supply and an increase in prices.

d) A consistent rise in supply due to technological advancements.

Answer: (c)

4. If consumer income suddenly decreases, which of the following is the likely effect on the demand curve?

   a) The demand curve shifts outward, indicating increased demand.

   b) The demand curve remains unchanged as income is held constant.

   c) The demand curve shifts inward, indicating decreased demand.

   d) The supply curve shifts to the right, resulting in lower prices.

   Answer: (c)

5. When technological advancements improve production efficiency, what is the expected impact on supply?

   a) The supply curve shifts left, leading to higher prices.

   b) The quantity supplied decreases, causing a surplus.

   c) The supply curve shifts right, leading to lower prices.

   d) The equilibrium price remains constant.

   Answer: (c)